

Education

Degree	Institution	CPI	Year
M.Tech (AI)	IIT Gandhinagar, Gujarat	8.00	2024 - Present
B.Tech (CSE)	IMS Engineering College, Ghaziabad	8.16	2019 - 2023

Internships

- **AI Product Engineer Intern**, AIResQ ClimSols, Research Park IIT Gandhinagar [Aug 2025 - Sept 2025]
 - o Developed a **scalable geospatial data pipeline** ingesting multi-source data (Satellite imagery, DEM, Rainfall) and generated **ground-truth flood data** for Bangalore using Sentinel-1 imagery.
 - o Built a **dataset automation tool** in Google Earth Engine and trained a **multi-input U-Net CNN** with **mid-layer rainfall fusion**, achieving **F1-score 0.84** (>87% pixel accuracy) on flooded pixels.
 - o Deployed a **Flask REST API** for real-time model predictions, enabling scalable inferencing and integration.
- **Software Engineer Intern**, XCoders Technologies, Roorkee [June 2022 - Aug 2022]
 - o Built an **Android app** with RecyclerView UI, **Firestore authentication**, and SQLite3 CRUD, enabling seamless login, registration, and offline data management.

Projects

- **AI-Powered EV Battery Safety System | Winner – AI Hackathon 2025, IIT Gandhinagar | [Link](#)** [Oct 2025]
 - o Built a **hybrid ML–RL safety intelligence framework** combining **ensemble learning** (RF + MLP) with a **rule-governed agent**, achieving **95+% detection accuracy** on **5.3M NASA samples** with **<10 ms latency**.
 - o Designed a **self-learning Q-learning safety agent** with **75% safety accuracy** and **6× state-space coverage gain** through **critical-scenario sampling** and **adaptive fine-tuning cycles**.
 - o Deployed an **interactive Streamlit dashboard** featuring a **Battery Health Index (BHI)** and **climate-adaptive safety thresholds** across **India’s five diverse climate zones** and **four seasons** for **predictive alerts** and **anomaly visualization**.
- **Smart Product Analytics Platform for Sellers (Full Stack + AI) | [Link](#)** [Aug 2025 - Sept 2025]
 - o Architected and deployed a full-stack **AI analytics platform** for Amazon product sentiment analysis, processing **25K+ reviews** with ASIN-based insights and cross-feature trends.
 - o Developed a **RAG-powered conversational AI** using **LoRA fine-tuned TinyLlama (1.1B)** with semantic search and fallback to pre-trained models under **8 GB AWS constraints**.
 - o Deployed on **AWS ECS Fargate** with **Docker**, **ALB load balancing**, and automated **CI/CD** via GitHub Actions and Terraform for scalable, reproducible infrastructure.

Achievements

- **Winner, National Level AI Hackathon 2025** held at IIT Gandhinagar, **Team Leader**
- LeetCode (**1840+** rating, **Top 7%**), HackerRank (5 star)
- Ranked **917/21,066 (Top 4%)** in LeetCode Biweekly Contest
- Secured **99.31 percentile** among **125K+ candidates** in **GATE(CS) 2024** examination
- **Finalist, Smart India Hackathon 2022, Team Leader**
- **Winner, Innovative Project Award 2021**, IMS Engineering College

Skill Summary

- **Programming:** C++, C, Python, Java; Strong fundamentals in Data Structures, Algorithms and OOP
- **AI/ML Engineering:** Proficient in Machine Learning and Deep Learning; experience across NLP (LLM fine-tuning, text analytics) and Computer Vision (classification, segmentation); skilled in model development, evaluation, and deployment
- **Web & Mobile Development:** HTML, CSS, Javascript; Familiar with Android
- **Tools & Technologies:** PyTorch, Sklearn, Pandas, NumPy, Matplotlib, OpenCV, Flask; SQL, Git, Docker

Certifications

- **IBM AI Engineering Specialization** | 9 courses | Coursera | **250+** hours
- **C++ Programming** | IIT Kanpur
- **Python 3 Specialization** | University of Michigan | Coursera | **60+** hours